
Host Website on Windows EC2 Instance Using IIS

[Edition 09]

[Last Update 230326]

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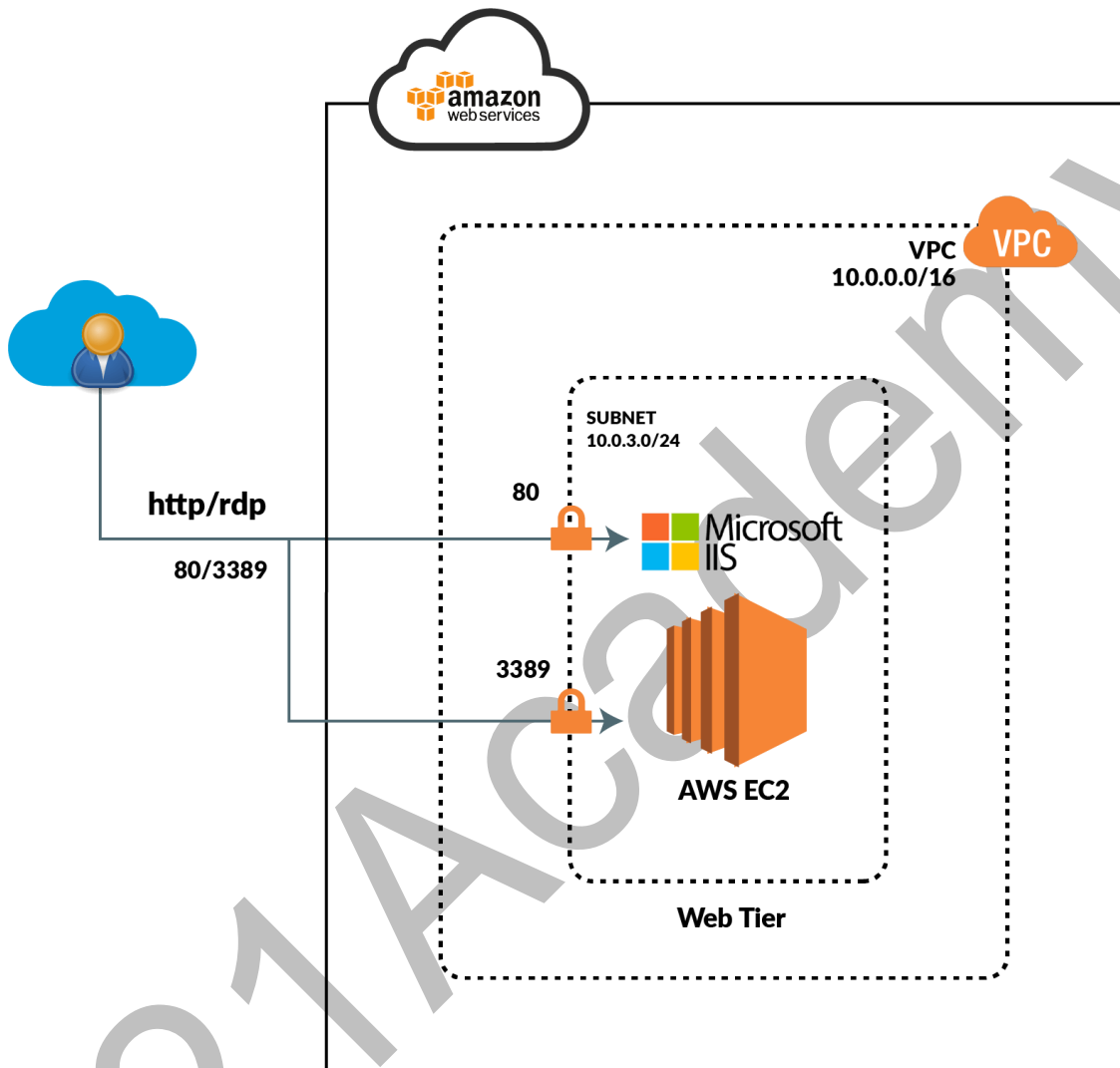
1 INTRODUCTION

Amazon EC2 presents a true virtual computing environment, allowing you to use web service interfaces to launch instances with a variety of operating systems, load them with your custom application environment, manage your network's access permissions, and run your image using as many or few systems as you desire.

To use Amazon EC2, you simply:

1. Select a pre-configured, templated Amazon Machine Image (AMI) to get up and running immediately, or create an AMI containing your applications, libraries, data, and associated configuration settings.
2. Configure security and network access on your Amazon EC2 instance.
3. Choose which instance type(s) you want, then start, terminate, and monitor as many instances of your AMI as needed, using the web service APIs or the variety of management tools provided.
4. Determine whether you want to run in multiple locations, utilize static IP endpoints, or attach persistent block storage to your instances.
5. Pay only for the resources that you actually consume, like instance-hours or data transfer.

Architecture Diagram:



This activity guide cover steps for:

1. Create Windows EC2 Instance
2. Connect to Windows EC2 Instance
3. Install Web Server (IIS) in Windows EC2 Instance
4. Access Webpage through Server Manager
5. Delete Resources

2 DOCUMENTATION LINKS

1. Amazon EC2
<https://aws.amazon.com/ec2/>
2. Features of Amazon EC2
<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html>
3. Amazon EC2 FAQs
<https://aws.amazon.com/ec2/faqs/>
4. Creating a Windows Instance Using Amazon EC2
https://docs.aws.amazon.com/AWSEC2/latest/WindowsGuide/EC2_GetStarted.html#ec2-launch-instance

3 PREREQUISITE

1. An AWS Account (Free or Paid).

K21Academy

4 CREATE A WINDOWS EC2 INSTANCE

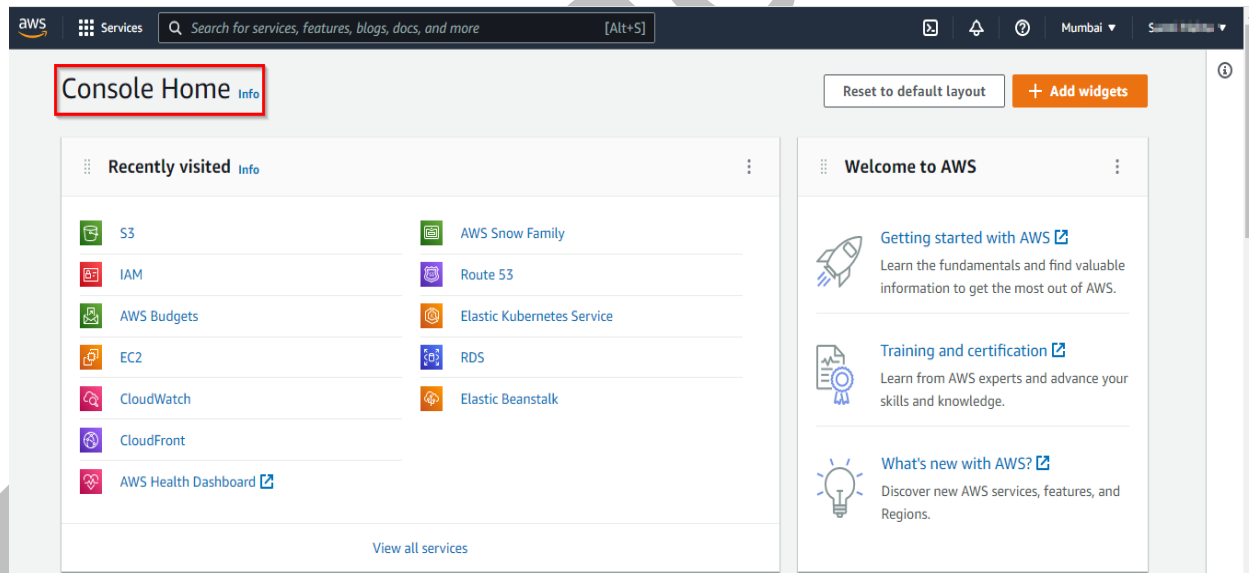
An **EC2 Instance** is a virtual server in Amazon's Elastic Compute Cloud (EC2) for running applications on the Amazon Web Services infrastructure. Amazon provides a variety of types of instances with different configurations of CPU, memory, storage, and networking resources to suit user needs. Each type is also available in two different sizes to address workload requirements.

4.1 Launch a Windows Instance

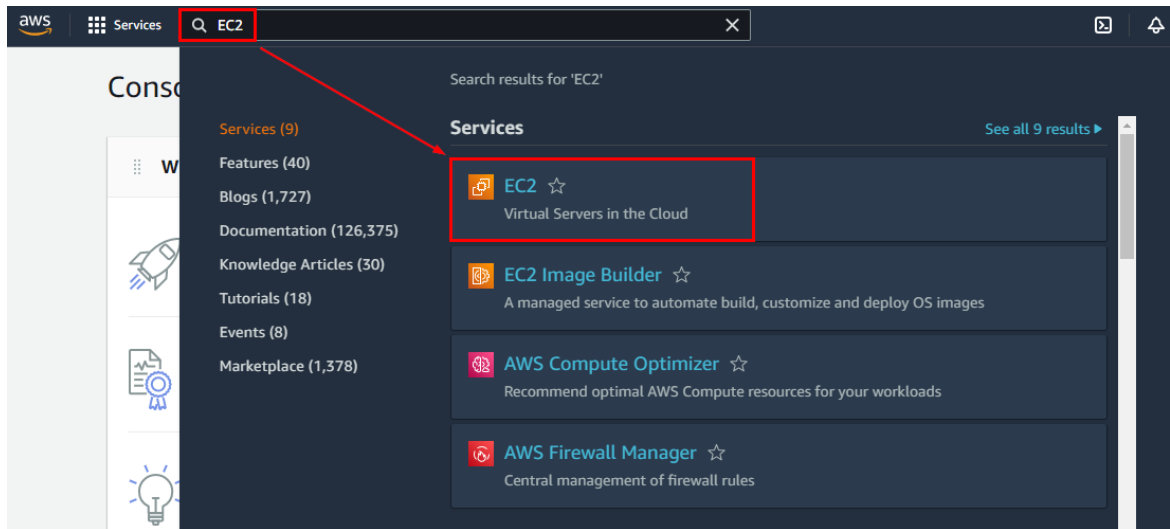
Note: In this section we will be providing detailed steps for creating a Windows Server 2019 Base

For this lab, we are using **N. Virginia Region**. You can select any region according to your choice.

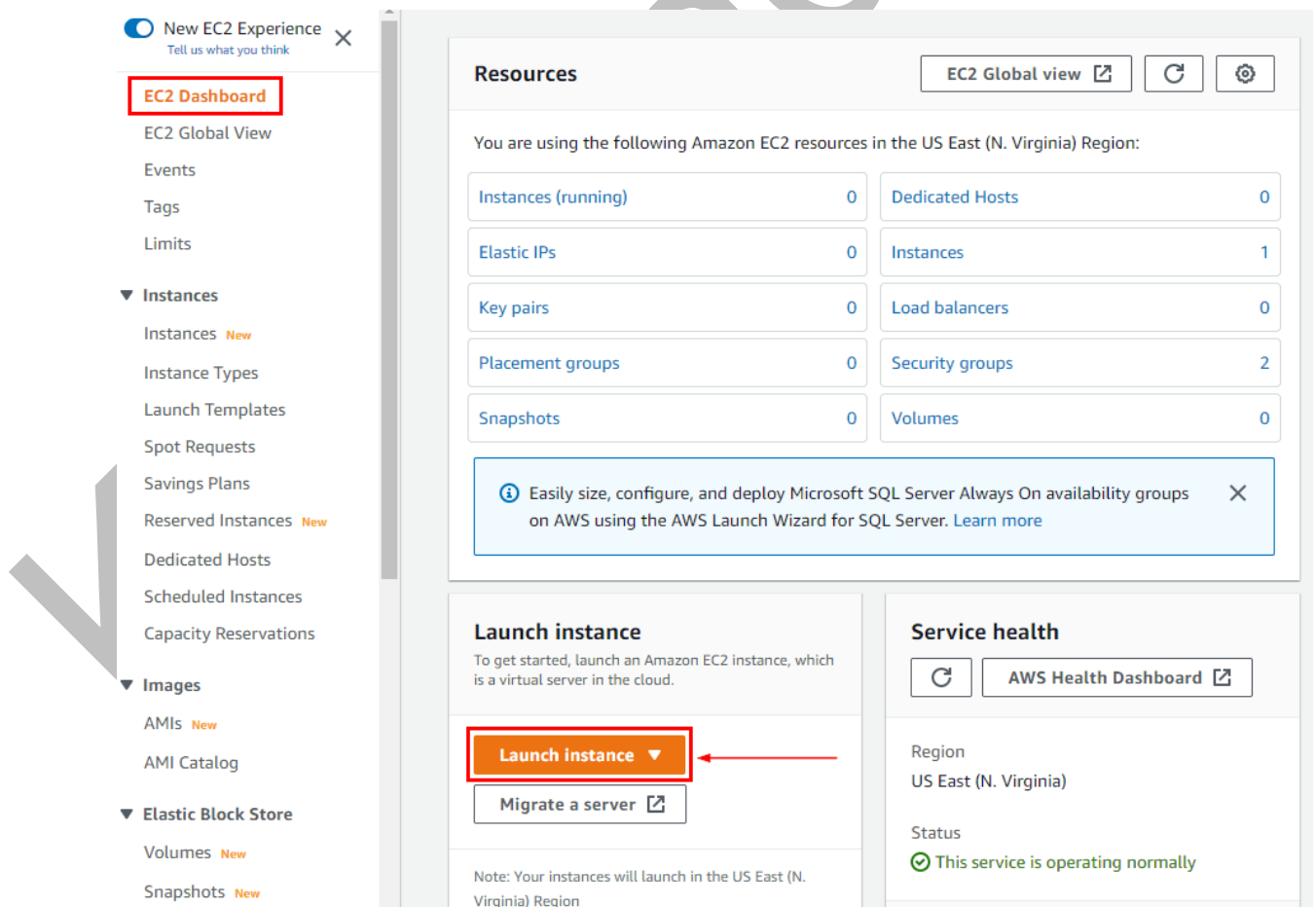
1. Log in to your **AWS Management Console**.



2. Search for **EC2** in the search box and click on it to open it.



3. Once you are at the EC2 Dashboard, click on **Launch** to launch an instance.



4. In the Name and Tags step you can add tags to an instance, here tags help you to enable categorize AWS resources in different ways, for example, by owner, environment, or purpose.

EC2 > Instances > Launch an instance

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

[Add additional tags](#)

Note: A **tag** is a label that you assign to an AWS resource. Tags enable you to categorize your AWS resources in different ways, for example, by purpose, owner, or environment. For example, you could define a set of tags for your account's Amazon EC2 instances that helps you track each instance's owner and stack level.

5. In Select **Windows** under QuickStart and Select **Microsoft Windows Server 2022 Base** AMI. You can also select other AMI as per your need but here we are launching a Windows Server, so we have to select the Windows Server 2022 Base AMI.

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

🔍 Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux macOS Ubuntu **Windows** Red Hat S

aws Mac ubuntu Microsoft Red Hat

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Microsoft Windows Server 2022 Base Free tier eligible ▼
ami-03cf1a25c0360a382 (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Microsoft Windows Server 2022 Full Locale English AMI provided by Amazon

Architecture	AMI ID	Verified provider
64-bit (x86)	ami-03cf1a25c0360a382	

- Select the **t2.micro** instance type, if you want you may select another instance type but they are chargeable so we choose the t2.micro instance type which is eligible for the free tier and limited resources.

Instance type Info

Instance type

t2.micro Free tier eligible

Family: t2 1 vCPU 1 GiB Memory

On-Demand Windows pricing: 0.0162 USD per Hour

On-Demand SUSE pricing: 0.0116 USD per Hour

On-Demand RHEL pricing: 0.0716 USD per Hour

On-Demand Linux pricing: 0.0116 USD per Hour

Compare instance types

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Create new key pair

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

Network settings Info Edit

Summary

Number of instances Info

1

Software Image (AMI)

Microsoft Windows Server 2022 ...read more

ami-0e38fa17744b2f6a5

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 30 GiB

Free tier: In your first year includes 750 hours of t2.micro for t3.micro in the

Cancel Launch instance

Note: T2 instances are a new low-cost, General Purpose instance type that are designed to provide a baseline level of CPU performance with the ability to burst above the baseline. T2 instances are engineered specifically for these use cases.

- Select an existing key pair or create a new one, we will Create a new one, enter the name of the Key-pair as Windows-Key and Create the Key Pair.

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select

Create new key pair

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

Create key pair ×

Key pairs allow you to connect to your instance securely.

Enter the name of the key pair below. When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#) 

Key pair name

Windows-Key

The name can include upto 255 ASCII characters. It can't include leading or trailing spaces.

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Cancel **Create key pair**

8. Since it is a Windows Instance, we will require **RDP** on port **3389** which is selected by default. Also, for accessing the web page we will add **HTTP** rule by clicking on **Add Rule**.

▼ **Network settings** Edit

Network
vpc-094a729d439e25fdd | Default VPC

Subnet
No preference (Default subnet in any availability zone)

Auto-assign public IP
Enable

Security groups (Firewall) [Info](#)
We'll create a new security group called 'launch-wizard-8' with the following rules:

☒ Allow RDP traffic from
Helps you connect to your instance Anywhere
0.0.0.0/0 ▼

☐ Allow HTTPs traffic from the internet
To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

Note: A **security group** acts as a virtual firewall for your instance to control inbound and outbound traffic. For, each security group, you add rules that control the inbound traffic to instances, and a separate set of rules that control the outbound traffic.

9. Now keep everything as it is and click on **Launch Instance**.

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

▼ **Configure storage** [Info](#) [Advanced](#)

1x GiB Root volume (Not encrypted)

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

The selected AMI contains more instance store volumes than the instance allows. Only the first 0 instance store volumes from the AMI will be accessible from the instance

0 x File systems [Edit](#)

► **Advanced details** [Info](#)

▼ **Summary**

Number of instances [Info](#)

Software Image (AMI)
Microsoft Windows Server 2022 ...read more
ami-0e38fa17744b2f6a5

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

Free tier: In your first year includes 750 hours of t2.micro for t3.micro in the

Cancel [Launch instance](#)

10. Now click on **View all Instances**.

EC2 > Instances > Launch an instance

✓ **Success**
Successfully initiated launch of instance (i-062c260586ec8a81f)

► Launch log

Next Steps

Create billing and free tier usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

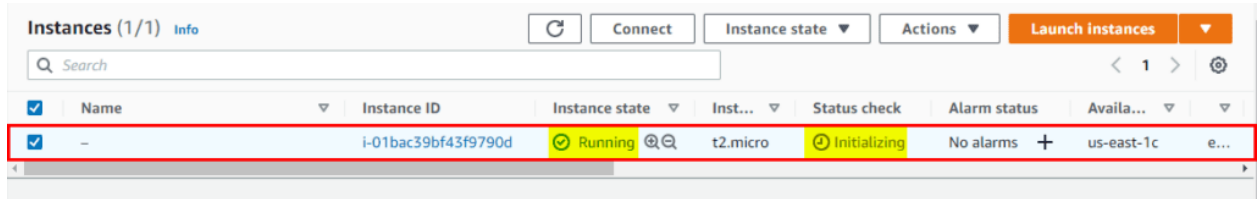
Connect an RDS database

Create a new RDS database

Learn more

[View all instances](#)

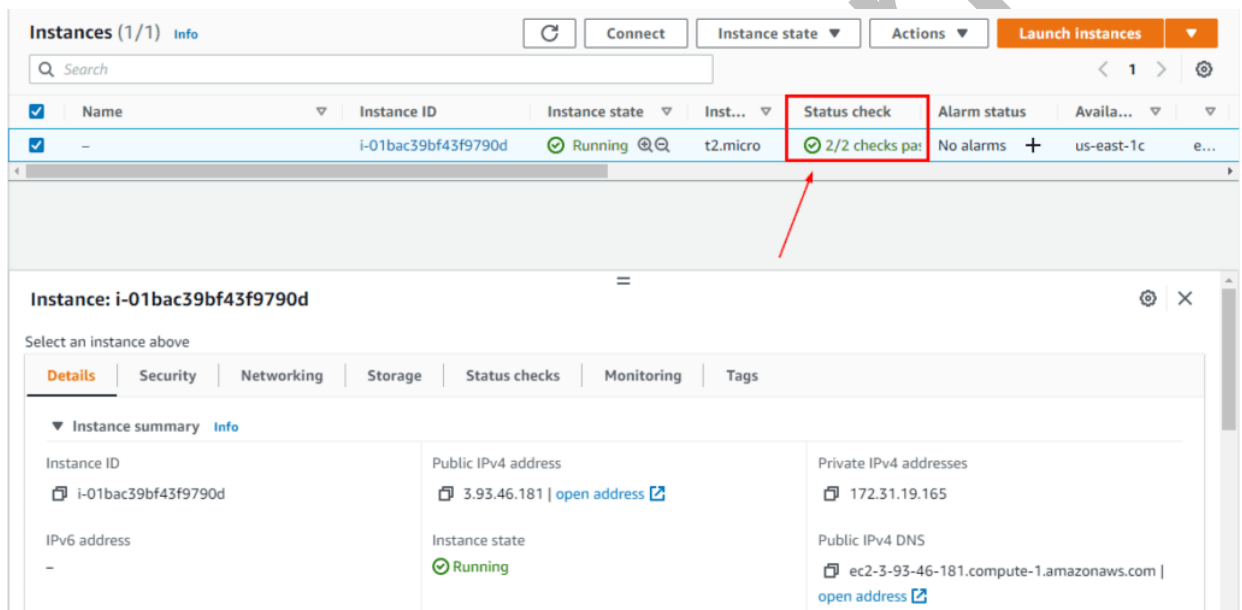
11. Here, you shall see your instance is launching and the Status check is **Initializing**, wait for some time.



The screenshot shows the AWS Management Console 'Instances' page. A single instance is listed with the following details:

Name	Instance ID	Instance state	Inst...	Status check	Alarm status	Availa...	e...
-	i-01bac39bf43f9790d	Running	t2.micro	Initializing	No alarms	us-east-1c	e...

12. Refresh and you shall see your instance is Up and Running, and the Status check has changed to **2/2 checks**.



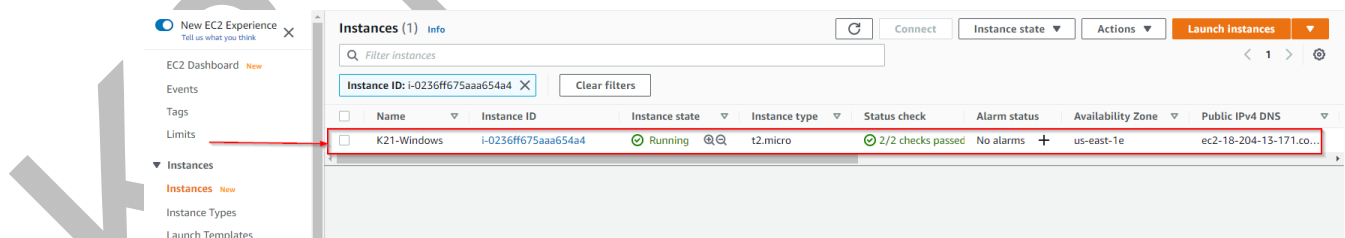
The screenshot shows the AWS Management Console 'Instances' page after a refresh. The instance is now in the 'Running' state, and the status check is '2/2 checks passed'. A red box highlights the 'Status check' column, and a red arrow points to the '2/2 checks passed' status.

Name	Instance ID	Instance state	Inst...	Status check	Alarm status	Availa...	e...
-	i-01bac39bf43f9790d	Running	t2.micro	2/2 checks passed	No alarms	us-east-1c	e...

Below the table, the 'Instance: i-01bac39bf43f9790d' details are shown. The 'Instance summary' tab is selected, displaying the following information:

- Instance ID: i-01bac39bf43f9790d
- Public IPv4 address: 3.93.46.181 | [open address](#)
- Private IPv4 addresses: 172.31.19.165
- IPv6 address: -
- Instance state: Running
- Public IPv4 DNS: ec2-3-93-46-181.compute-1.amazonaws.com | [open address](#)

13. Now we have successfully created our **Windows EC2 Instance**.



The screenshot shows the AWS Management Console 'Instances' page with a sidebar on the left. The 'Instances' section is selected in the sidebar. The main table lists the following instance:

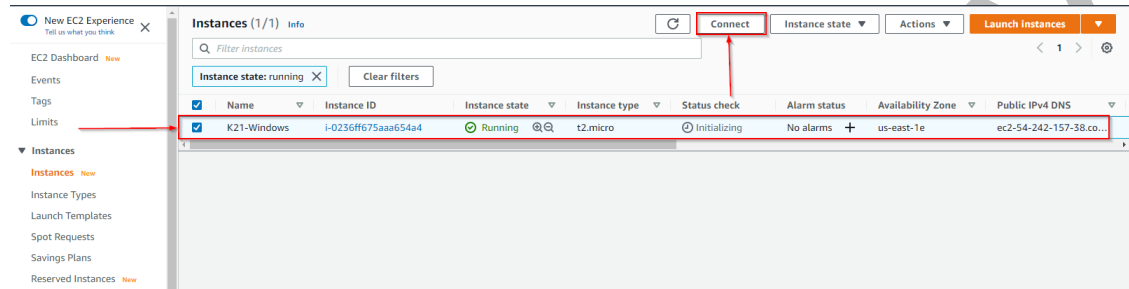
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
K21-Windows	i-0236ff675aaa654a4	Running	t2.micro	2/2 checks passed	No alarms	us-east-1e	ec2-18-204-13-171.co...

Thus we have successfully created the Windows Instance

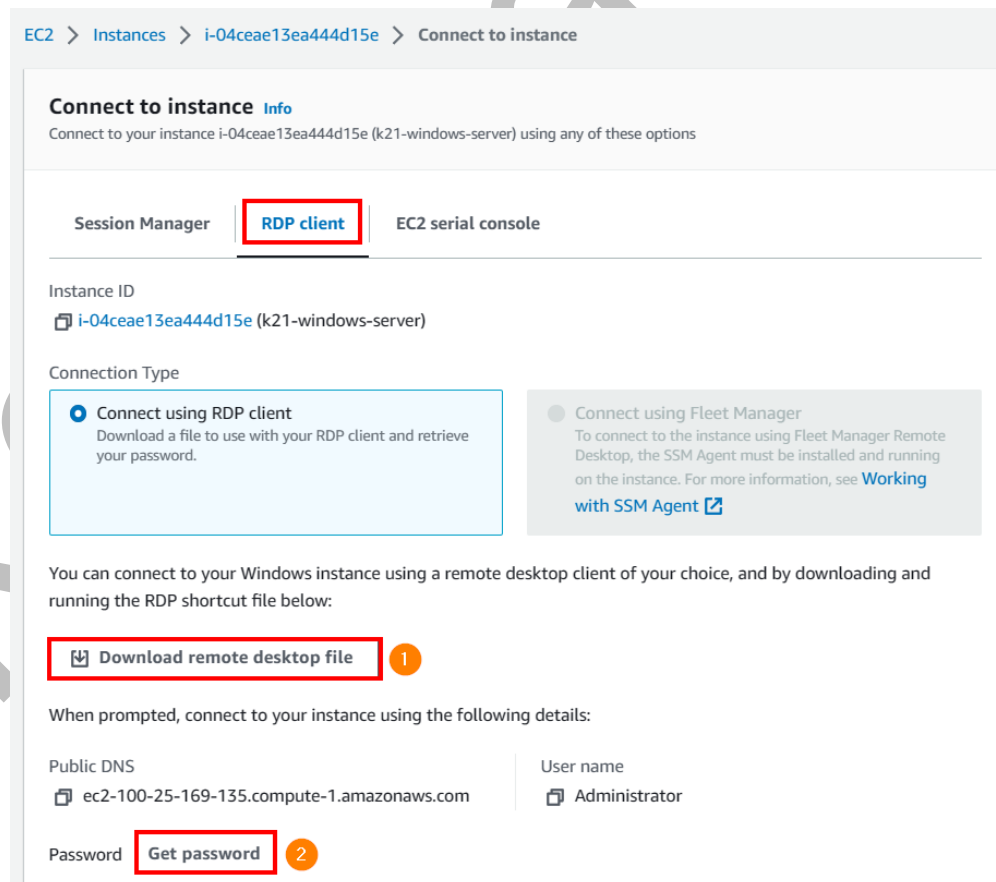
4.2 Connect to the Windows Instance

Note: In this section we will be providing detailed steps for connecting to our Windows Instance using Remote Desktop Connection.

1. Select the **Windows Instance** we just created and click on **Connect**



2. Here we have to select the **RDP (Remote desktop protocol)** Client and then Download the RDP File and save it somewhere safe then, and we need a password to access the RDP file, so click on **Get Password**.



3. Once you click on the **Get Password**, click on **Upload private key file** and select the **key pair** we just created at the time of instance launching.

EC2 > Instances > i-04ceae13ea444d15e > Get Windows password

Get Windows password [Info](#)

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID
i-04ceae13ea444d15e (k21-windows-server)

Key pair associated with this instance
windows-key

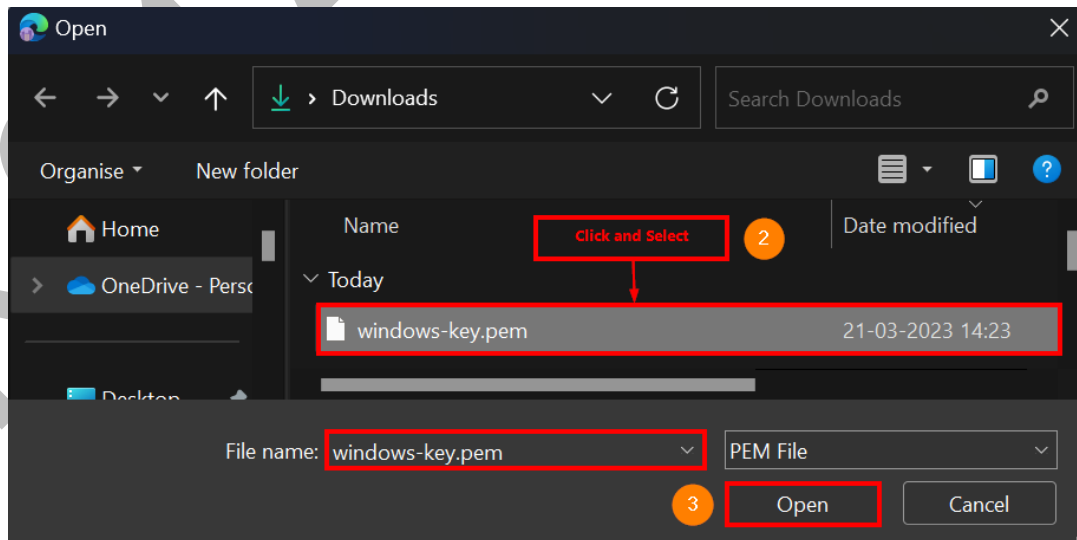
Private key
Either upload your private key file or copy and paste its contents into the field below.

Upload private key file 1

Private key contents - optional

Private key contents

Cancel Decrypt password



4. Click on **Decrypt Password**

EC2 > Instances > i-04ceae13ea444d15e > Get Windows password

Get Windows password [Info](#)

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID
i-04ceae13ea444d15e (k21-windows-server)

Key pair associated with this instance
windows-key

Private key
Either upload your private key file or copy and paste its contents into the field below.

☒ windows-key.pem
1.674KB

Private key contents - optional

```
-----BEGIN RSA PRIVATE KEY-----
MIIIEogIBAACAQEAq69PJvp48KQWXkKIC/CYfJ0itZDend/WyyLJ3SSlo5uMrRiL
3WZq/sf0NEey6jpK9m+oLRmlZEcmMjGZI0FseVZFq2uLEoAX0JrifNcVBObg/jYF
aiP8AR0ww51gFa+dw8R/Gg1qNb1QA2ln1YB9lVKG6keRETBmnfFWdSyZqSy4nLey
fxJElmbZpTmlHaDuRQeNRPOpSeawFbuy/aHW1o3TpenF6EMfbr6ygEXZc7v/cNiD
Ls6b1uoqQwXauMmm4PvRpfMX53wISqZyXsW9GWwMVDdGWUxBpKQFoHsyHRM67G7F
HhLi35nZLsC2FOvwQJ1aBLoZP8eeyDbym2aSHQIDAQABAoIBACGKHvIrZyMDu0bw
0mkauH9npzc3NPY8m1aV1j+4gA9lPOc362L2NiLN5zwrYZ12g9x+1ajx4KilaaXW
-----
```

5. After this you will get the **decrypted password** for the instance.

EC2 > Instances > i-0236ff675aaa654a4 > Connect to instance

Connect to instance [Info](#)

Connect to your instance i-0236ff675aaa654a4 (K21-Windows) using any of these options

Session Manager | **RDP client**

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following details:

Public DNS
ec2-54-242-157-38.compute-1.amazonaws.com

User name
Administrator

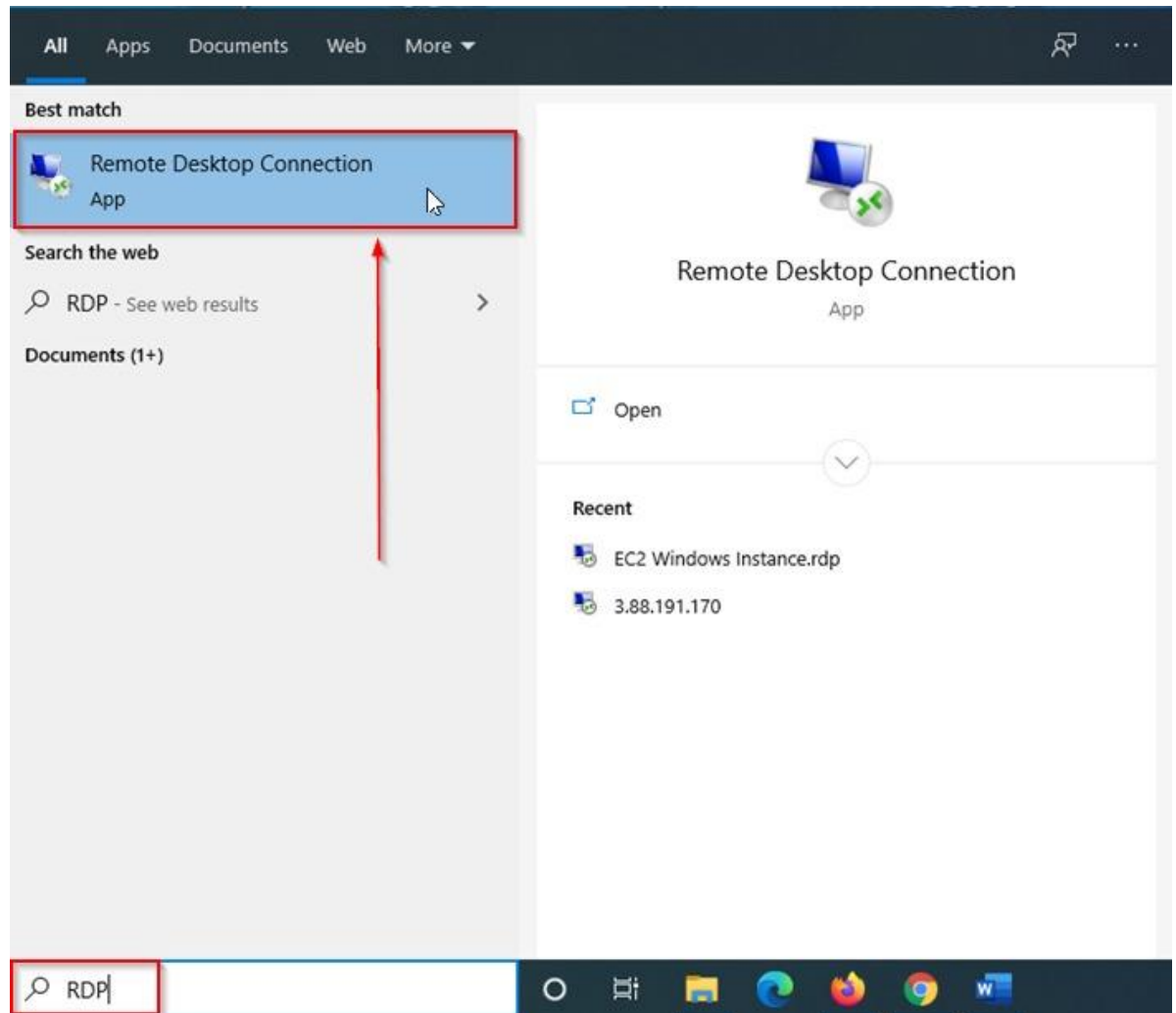
Password
7V.ISlu5UAV)@Lyx3=g\$iZNdGgc3MHnR

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

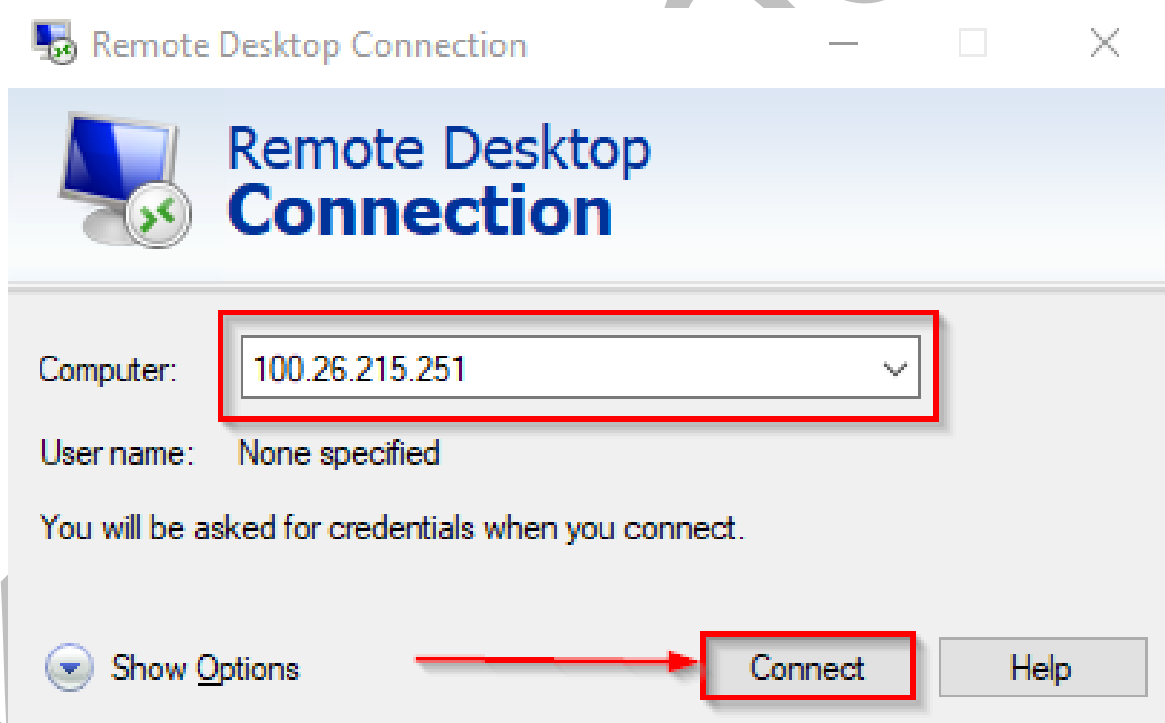
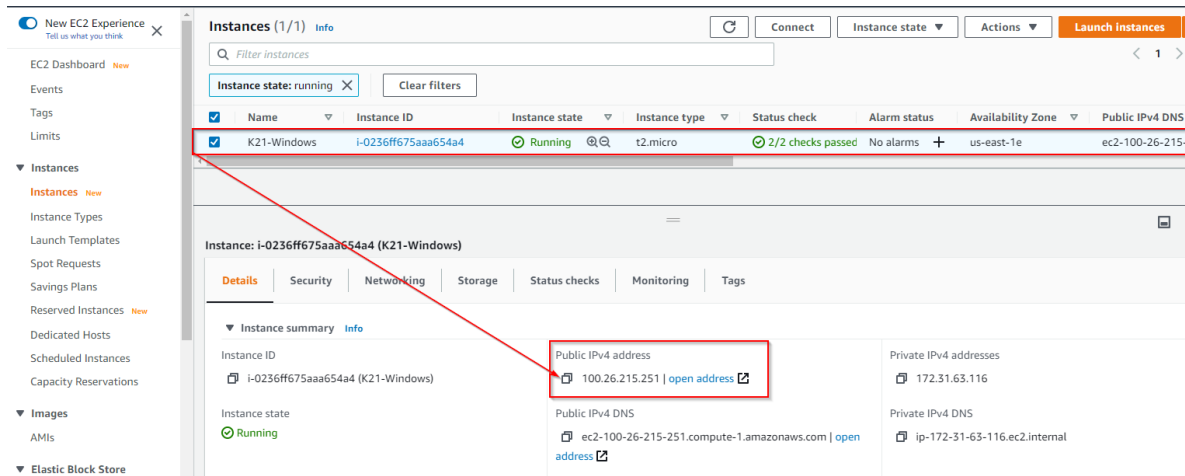
Cancel

Note: Make a note of this **Username and Password** as we will be using this password for connecting to the Windows Instance.

6. To connect to the instance again, search for **RDP** in the Start menu and open it.

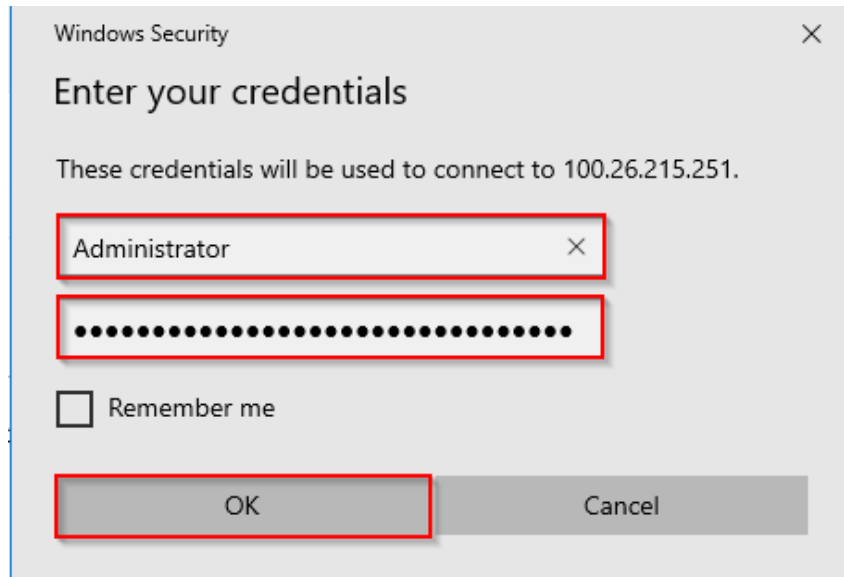


7. Now copy the **Public IP** of the Windows Instance and paste it and click on **Connect**

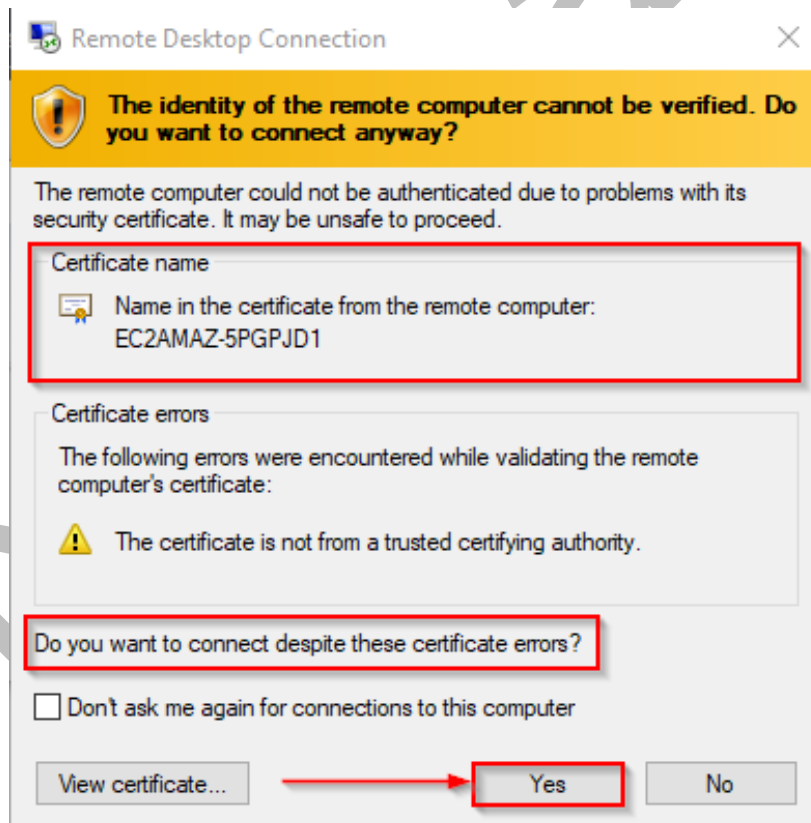


Note: You can also open Remote Desktop File from Downloads folder on your PC for launching the Windows instance.

8. Now enter the **Username** (**Administrator** by default) and the **Password** that you had decrypted earlier.



9. Click on **Yes**.



10. You will be logged in to the **Windows Instance**

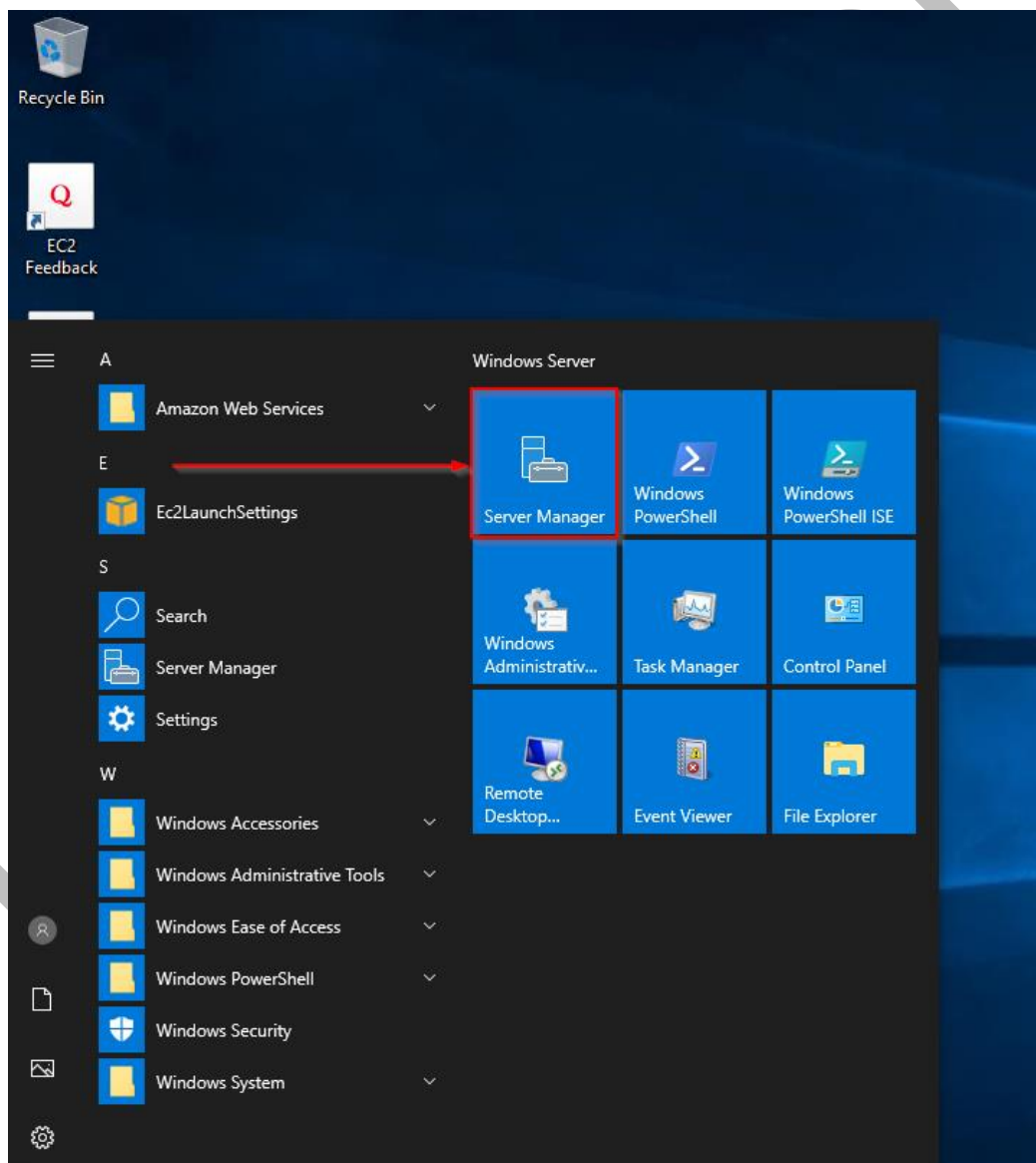


Thus, we have successfully logged in to the Windows Instance

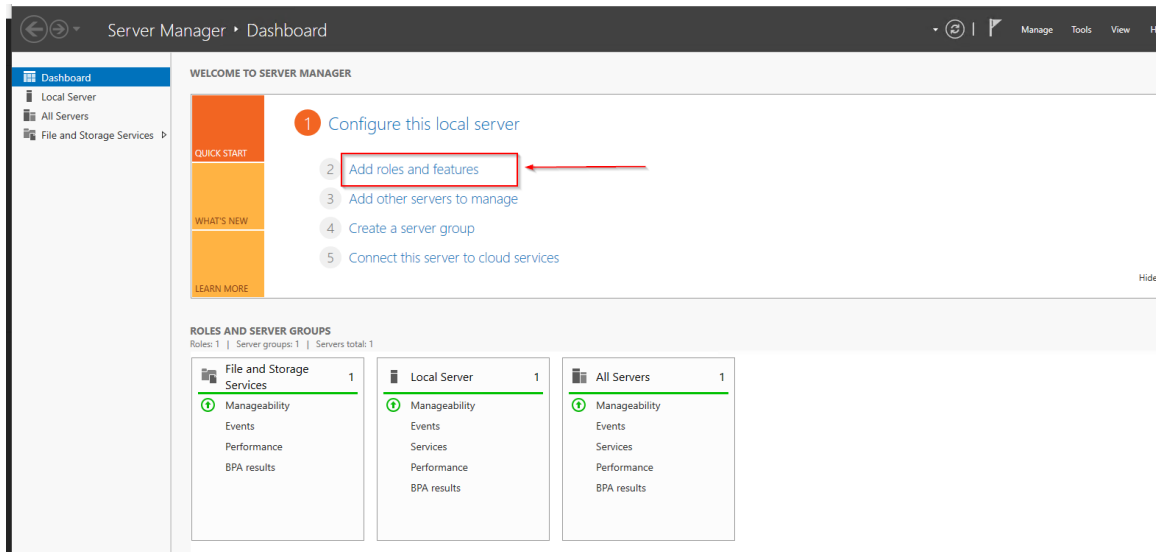
5 INSTALL WEB SERVER (IIS) ON WINDOWS EC2 INSTANCE

Internet Information Services (IIS, formerly Internet Information Server) is an extensible web **server** software created by **Microsoft** for use with the **Windows** NT family. IIS is used to host ASP.NET web **applications** and static websites. It can also be used as an FTP server, host WCF services, and be extended to host web applications built on other platforms such as PHP. There are built-in authentication options such as Basic, ASP.NET, and Windows auth.

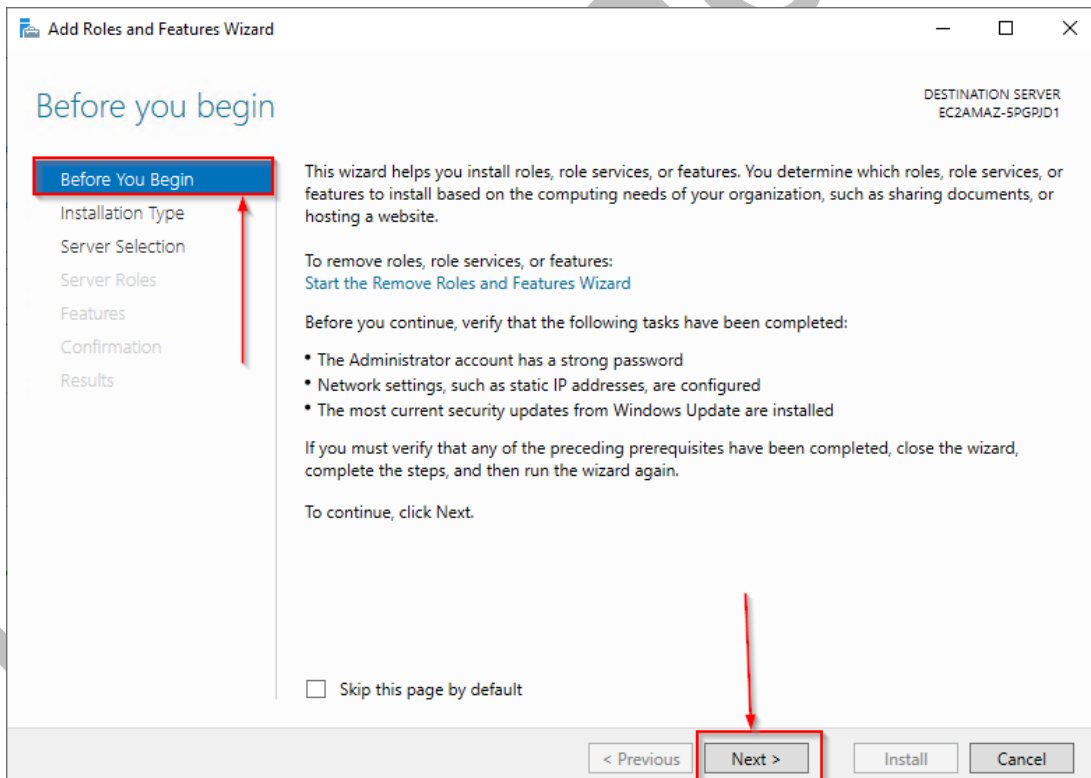
1. Once logged in to the Windows Instance, click on **Start button** and then click on **Server Manager**.



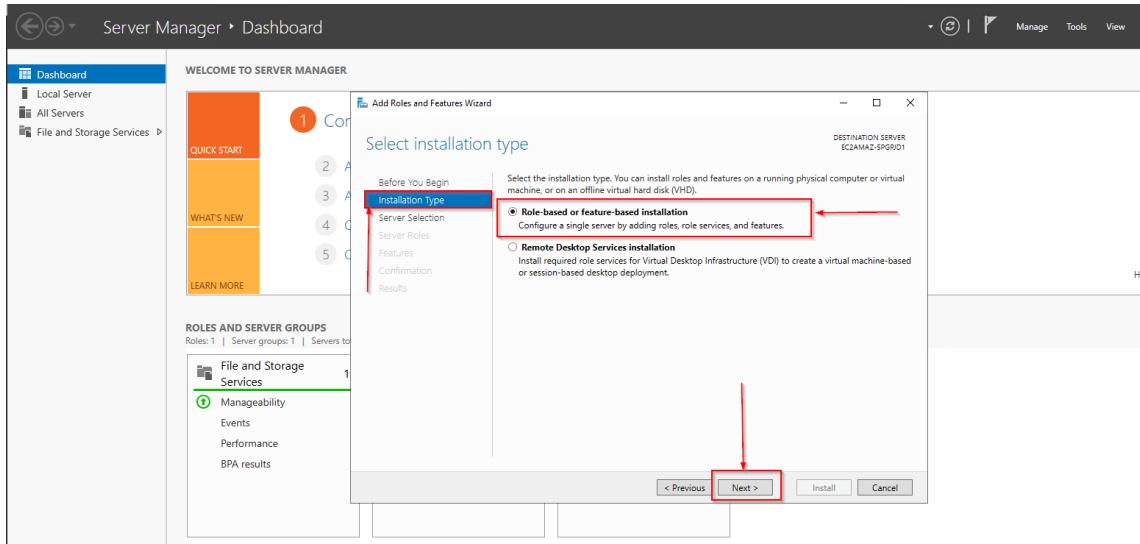
2. Click on **Add roles and features**.



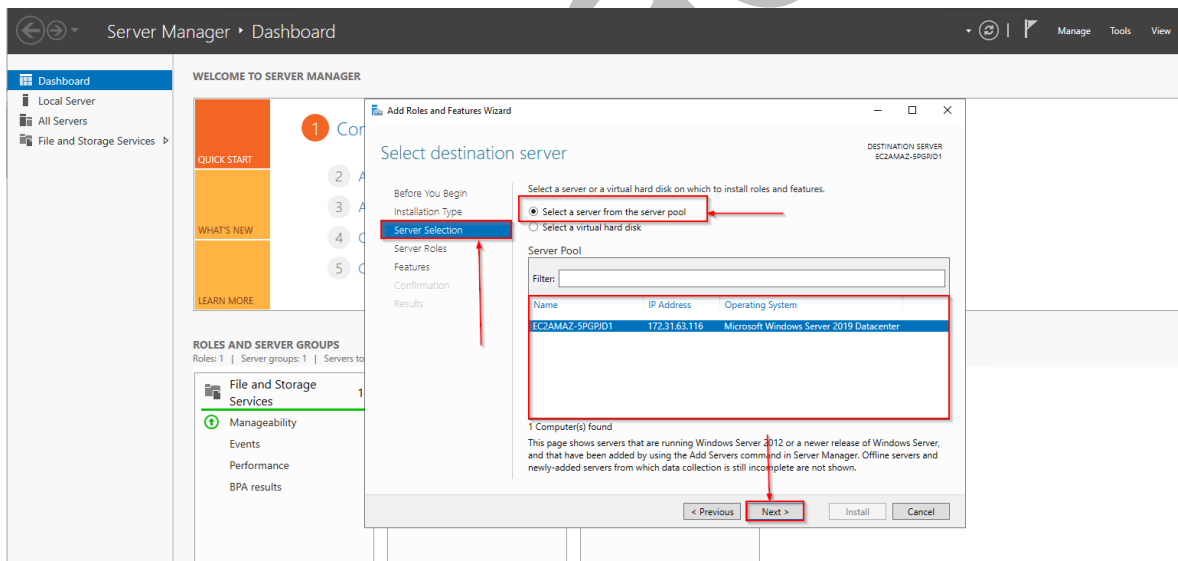
3. Click on **Next**.



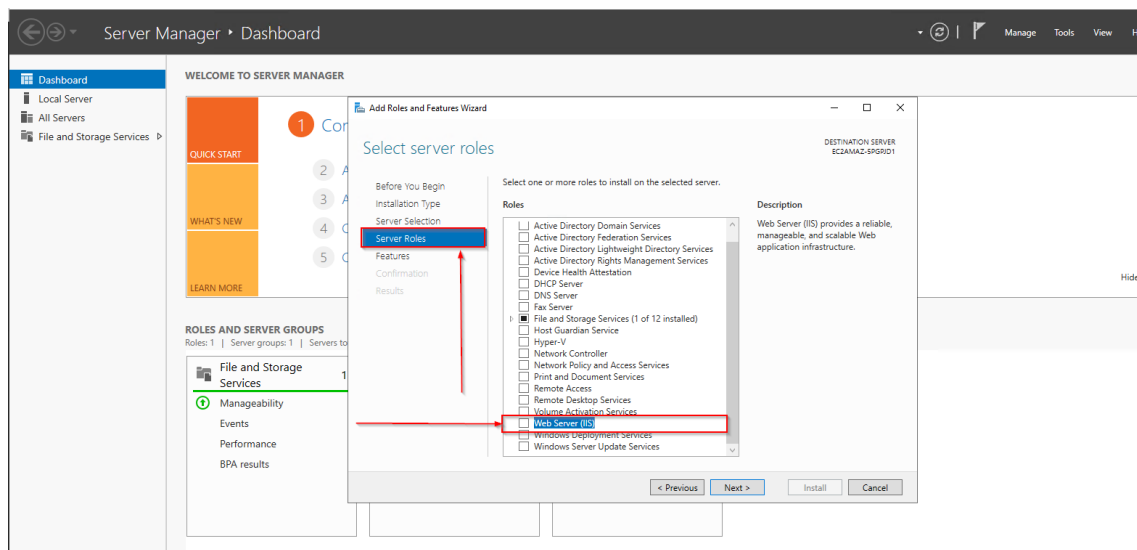
4. Click on **Role based or feature based Installation** and then click on **Next**



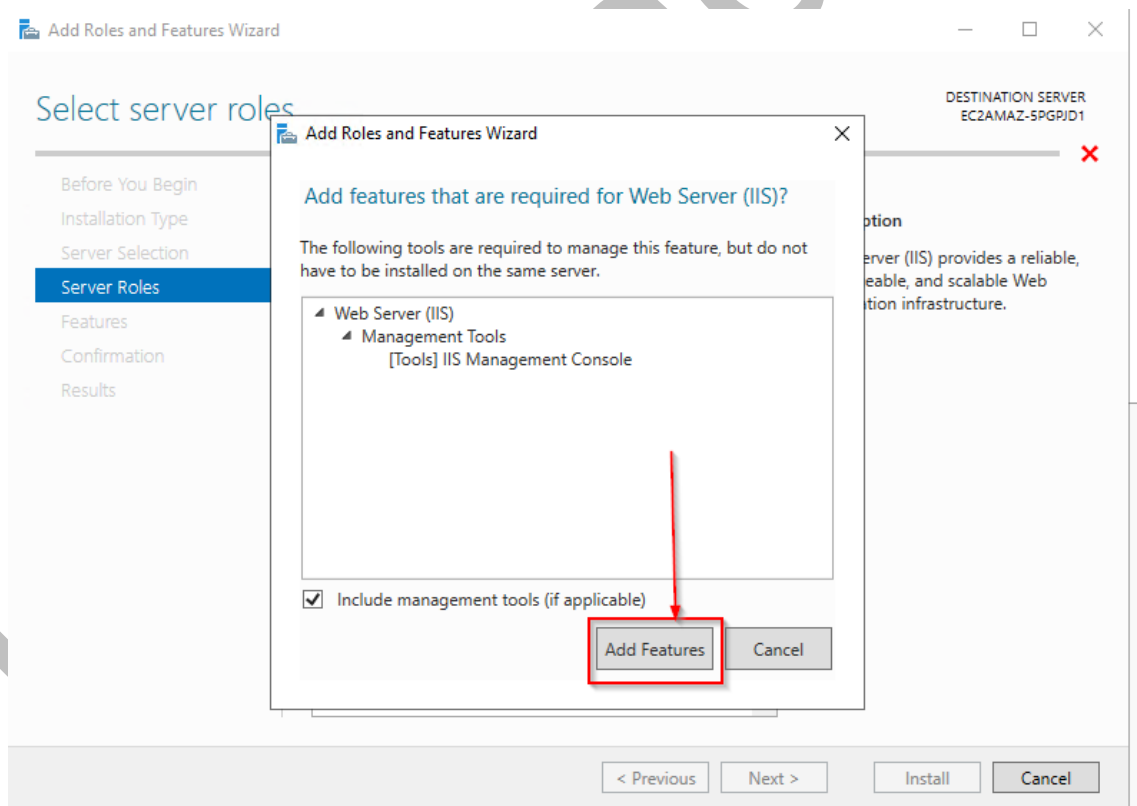
5. Keep everything **default** and click on **Next**



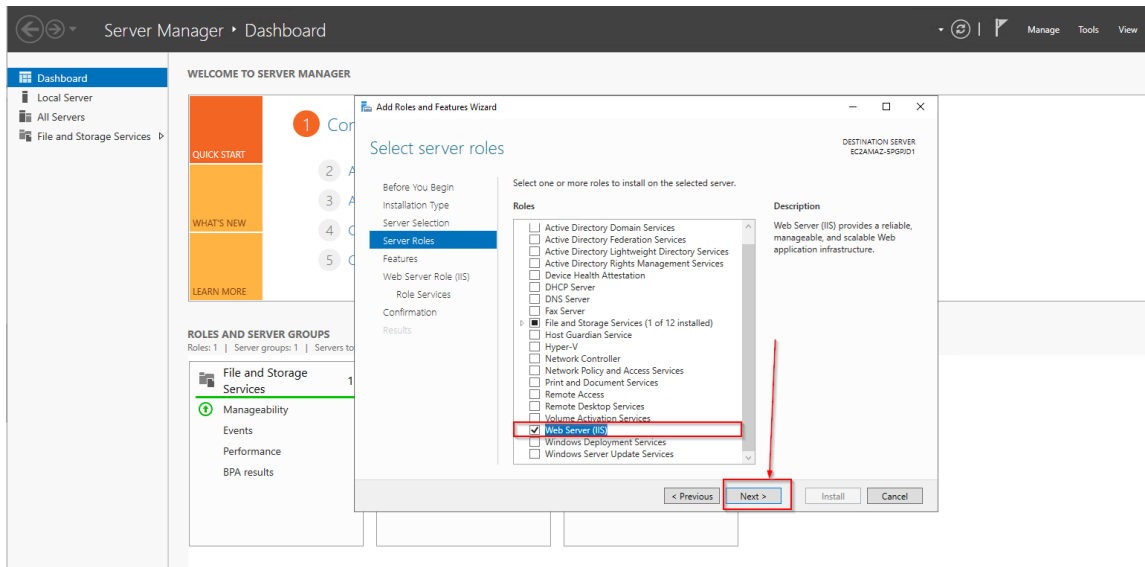
6. Scroll down a bit and click on **Web Server (IIS)**



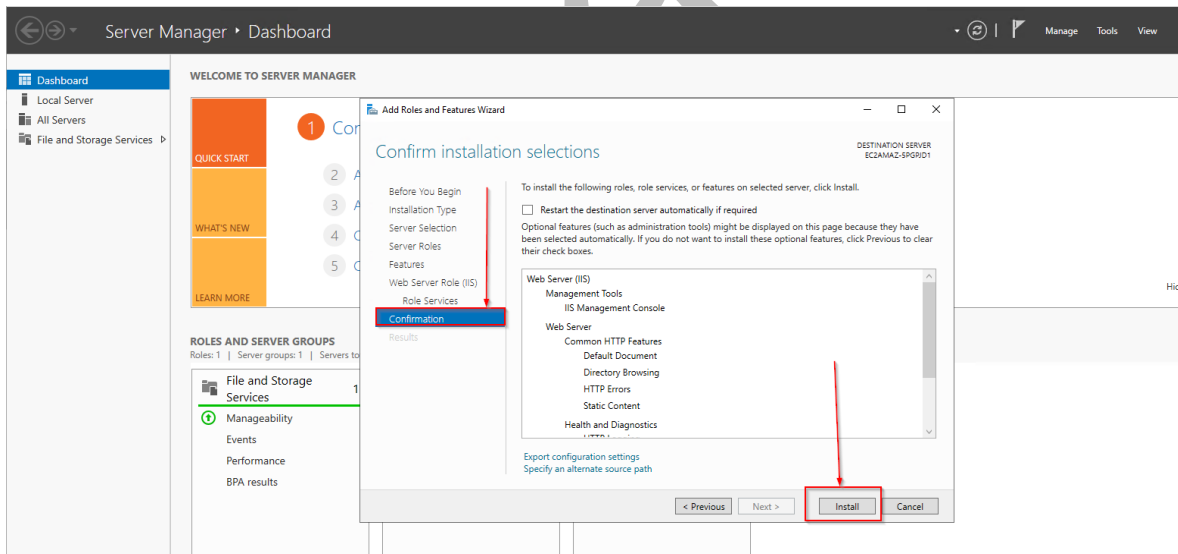
7. As you click on **Web Server (IIS)**, a small dialog box will open. Click on **Add Features**



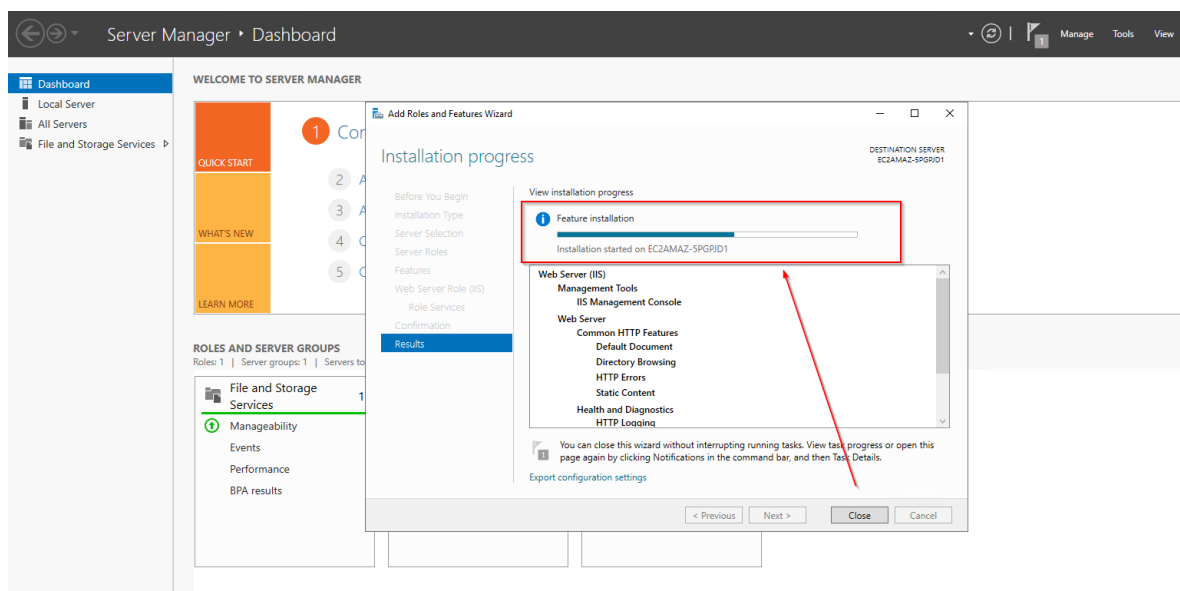
8. Now click on **Next** again.



9. Now keep clicking on **Next** till you reach **Install** dialog box.

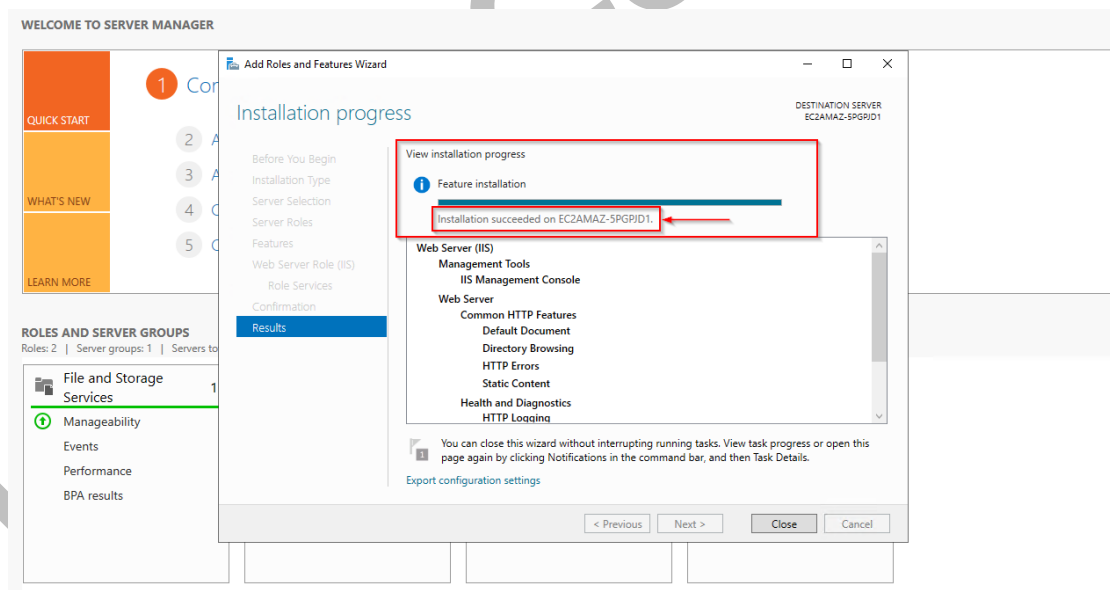


10. The **installation** will start as shown below.



Note: The feature installation will take some time (**approx. 5 minutes**) to complete. Please be patient.

11. The installation is **complete** now.

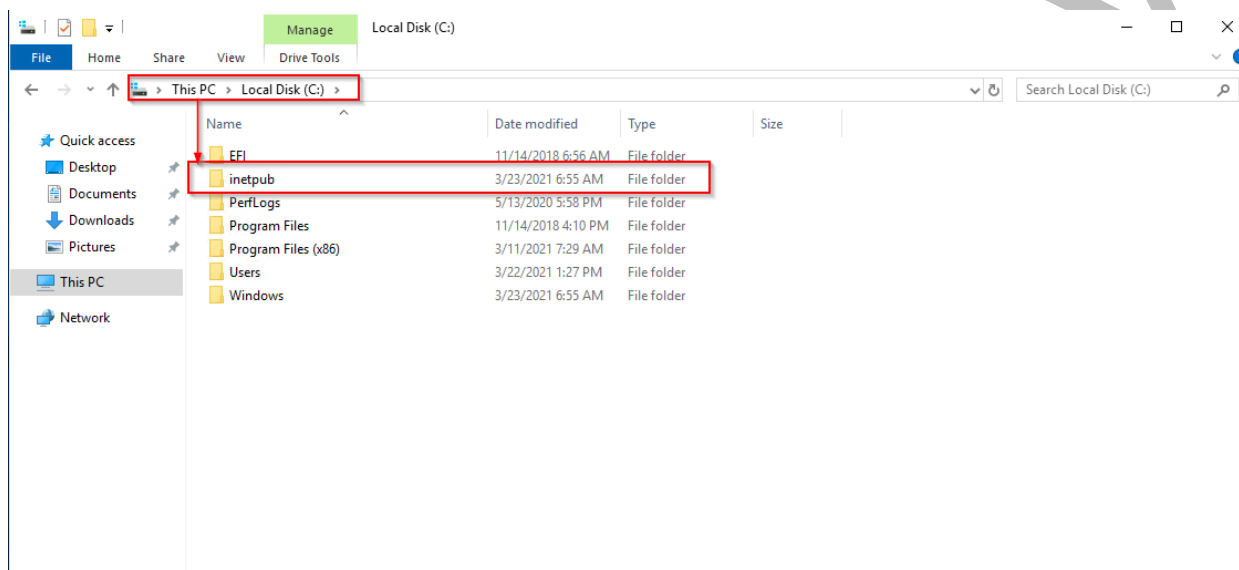


Thus we have successfully installed IIS on our Windows Instance.

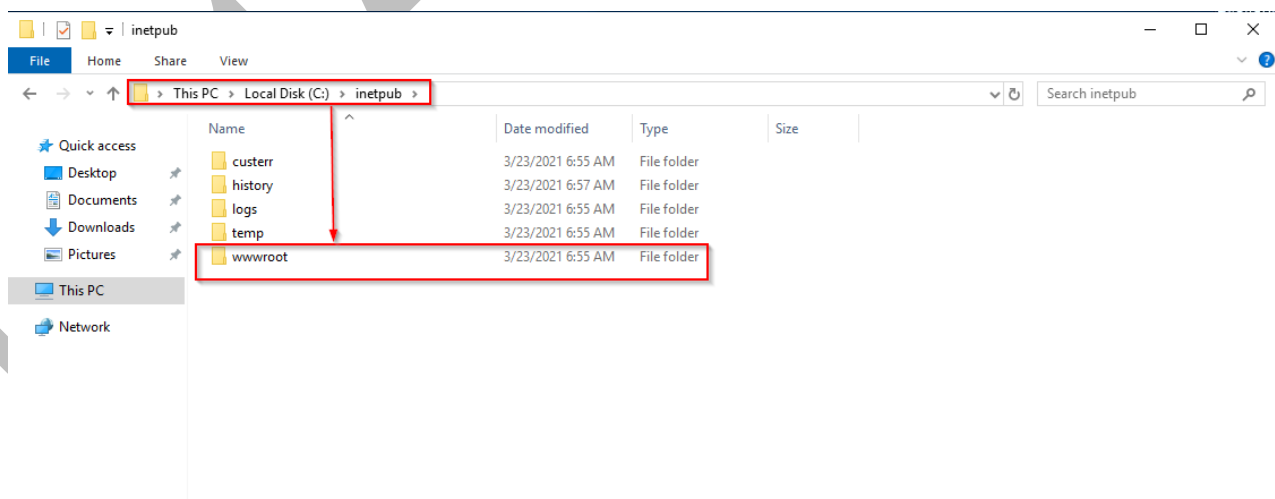
6 ACCESS WEB PAGE THROUGH SERVER MANAGER

In this section we will be providing detailed steps as to how you can access the web page after you have installed IIS on your Windows virtual machine

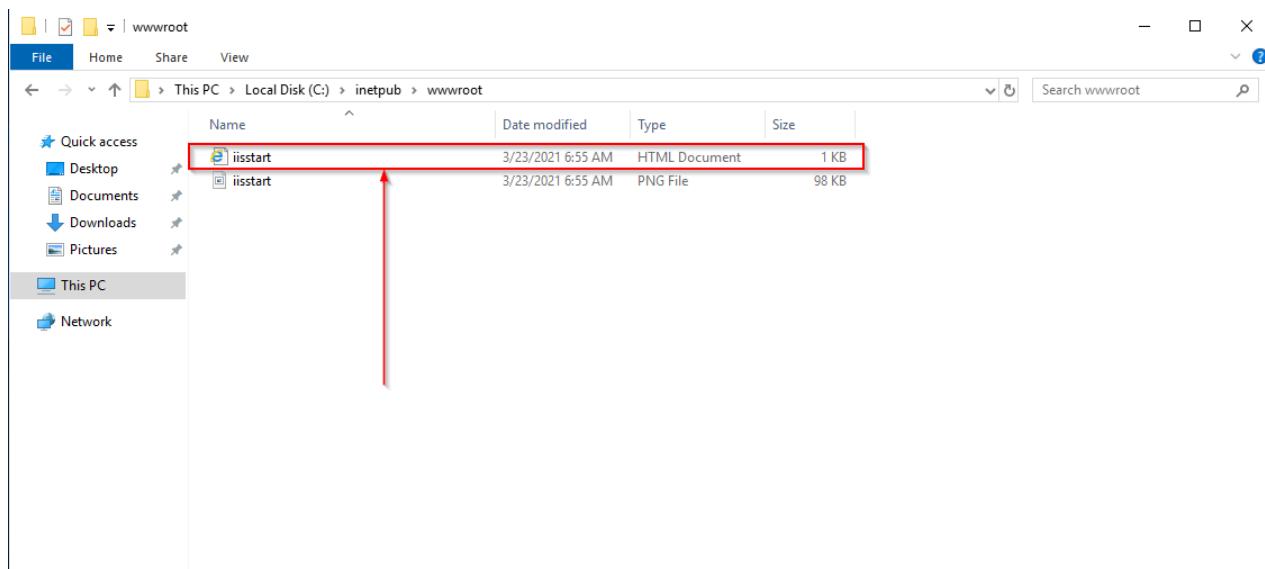
1. Navigate to the **local disk** of you Windows Virtual machine. You will see a folder by the name **inetpub** created in your local disk. Double click to open it.



2. Once you have opened it, now double click on **wwwroot** folder to open it.



3. You will find an HTML document inside by the name **iisstart**. This will be the **home page** of your web server.



4. Right click on this file and open with **Notepad**. It contains a HTML code as shown below.

```
iisstart - Notepad
File Edit Format View Help
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1" />
<title>IIS Windows Server</title>
<style type="text/css">
<!--
body {
    color:#000000;
    background-color:#0072C6;
    margin:0;
}

#container {
    margin-left:auto;
    margin-right:auto;
    text-align:center;
}

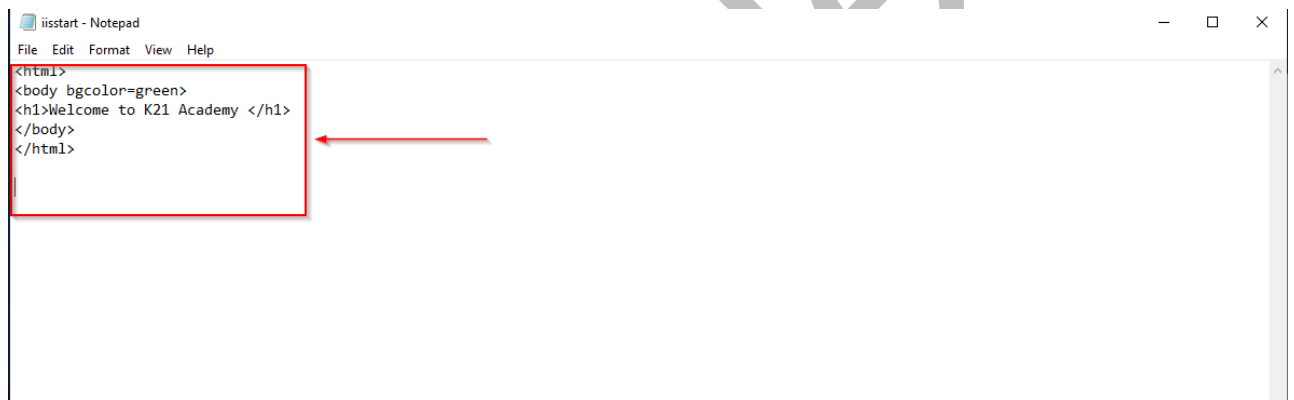
a img {
    border:none;
}

-->
</style>
</head>
<body>
<div id="container">
<a href="http://go.microsoft.com/fwlink/?linkid=66138&clid=0x409"></a>
```

5. Now to make our web page look a bit attractive. We will edit this HTML code by erasing everything in it and write our own HTML code and **Save** it.

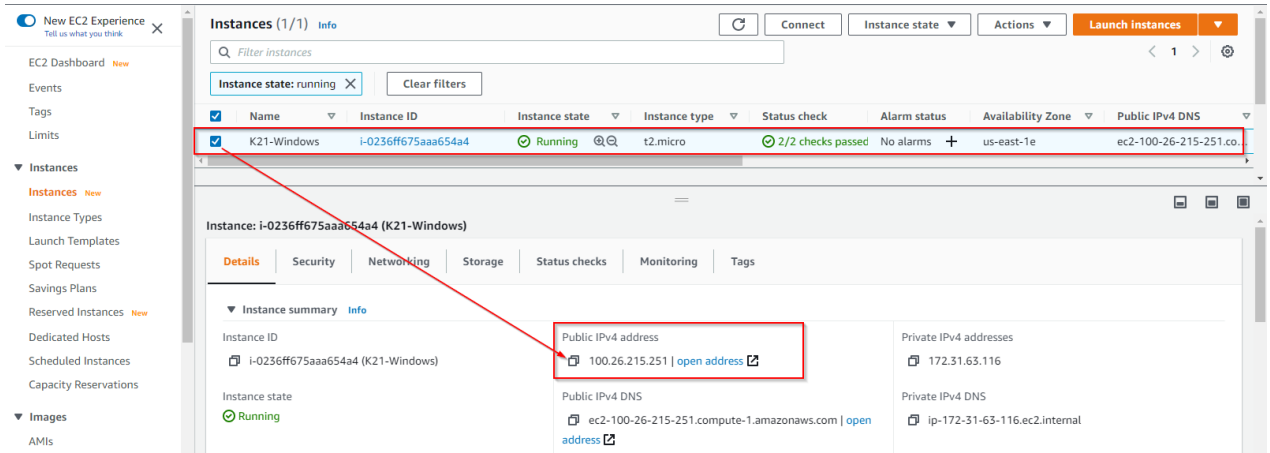
Note: If you don't want to edit the existing **HTML code** and want to keep it as it is you can skip this step and follow the next step

```
<html>
<body bgcolor=green>
<h1>Welcome to K21 Academy </h1>
</body>
</html>
```

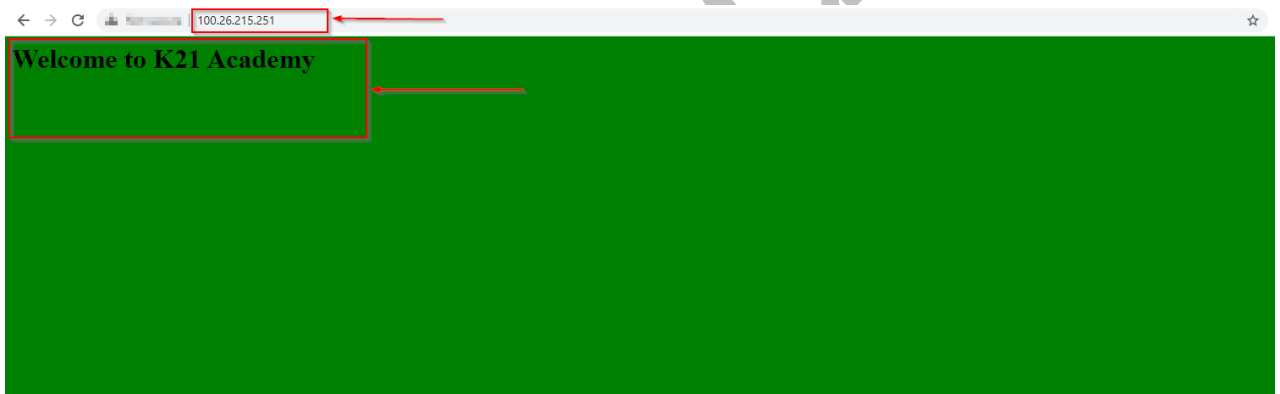


Note: Don't forget to save it by clicking on **File** at the top left corner and then click on **Save**. If you don't save the changes will not appear on the web page

6. Switch to the **EC2 console**, select the **Windows Instance** and copy its **Public IP**.



7. Now try hitting this **Public IP** of the **Windows Instance** in your Browser.



Thus we have successfully accessed the Webpage

7 DELETE RSOUCES

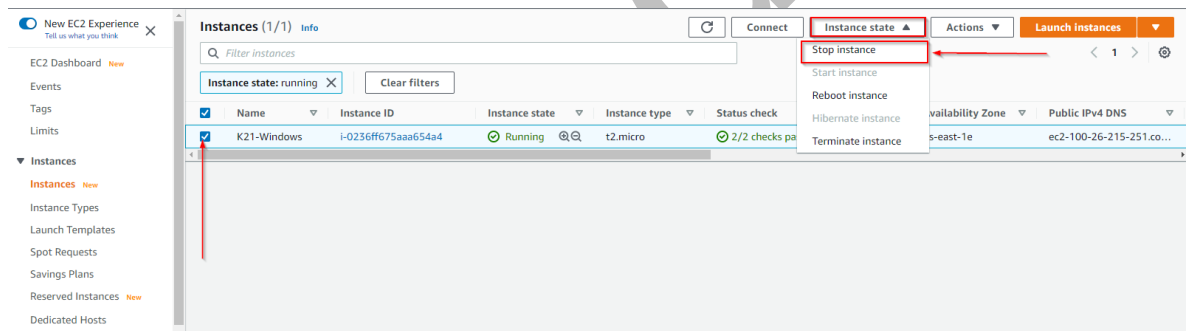
In this section, we will delete all the resources that we used in the lab as keeping them running will charge us. So, all the resources must be deleted once you do not require them in order to be in free tier limit.

Note: Kindly take screenshot of the Lab you have created and post it in our WhatsApp group so that we can see you are making progress. Also, it will inspire others to perform this LAB.

7.1 Stop the Windows Instance

Note: This section will cover steps to stop the windows instance

1. Go to your Instance and select the Windows Instance. Click on **Instance State** and click on **Stop Instance**.



2. Click on **Stop** to stop the Instance

Stop instance? ×

Instance IDs

 i-0236ff675aaa654a4 (K21-Windows)

To confirm that you want to stop the instance, choose the *Stop* button below.

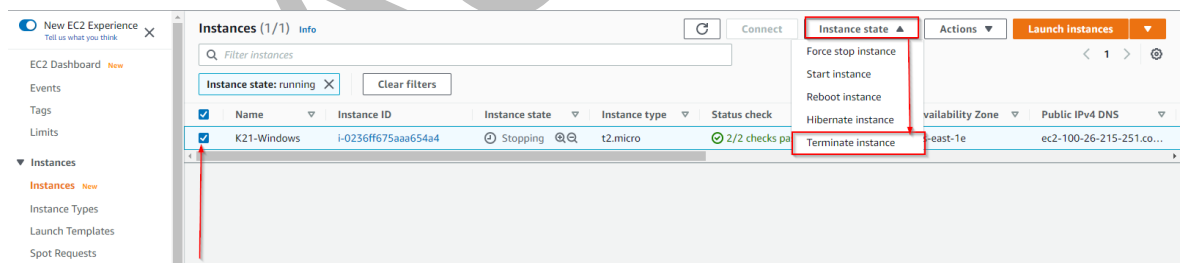
Cancel **Stop**

Thus, we have successfully stopped the Instance.

7.2 Terminate the Windows Instance

Note: This section will cover steps to terminate the windows instance

1. Go to your Instance and select the Windows Instance. Click on **Instance State** and click on **Terminate Instance**.



2. Click on **Terminate** to terminate the Instance

Terminate instance?

×

Instance IDs

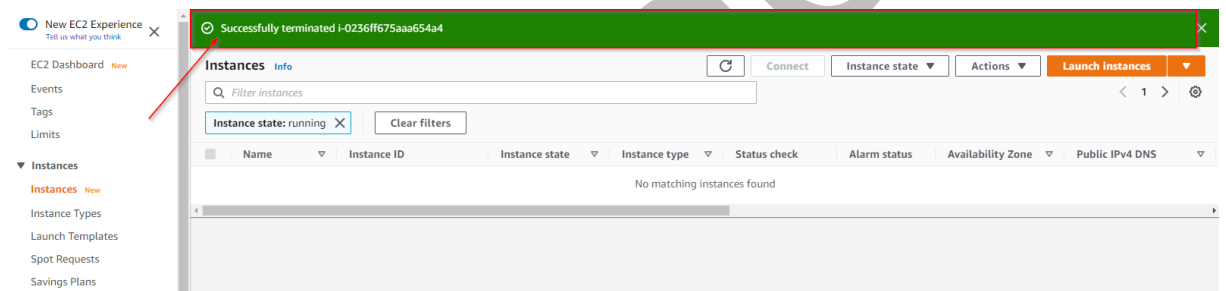
 i-0236ff675aaa654a4 (K21-Windows)

To confirm that you want to terminate the instance, choose the *terminate* button below.
Terminating the instance cannot be undone

Cancel

Terminate

3. The Instance will be **deleted**.



Thus we have successfully deleted the Instance

8 TROUBLESHOOTING

Note: In this section, we will document any issues and their fixes that may arise while performing this lab. If you encounter any issues, please let us know so that we can update this section with the relevant information and help others who may face similar problems.

K21Academy

9 SUMMARY

This activity guide cover steps for:

1. Create Windows EC2 Instance
2. Connect to Windows EC2 Instance
3. Install Web Server (IIS) in Windows EC2 Instance
4. Access Webpage through Server Manager
5. Delete Resources